Anastasios Tsolakis

Robotics and Control Engineer
Patras, Greece

+30 697 5635728
tas.tsolakis@gmail.com
Portfolio
LinkedIn
Scholar
GitHub

Profile

Robotics and Control Engineer (PhD, TU Delft) with expertise in dynamics, motion planning, fault-tolerant control, and ROS2-based system integration. Strong mechanical engineering foundation (kinematics, dynamics, mechanisms, design, CAD/CAE) and hands-on deployment of autonomous platforms (ASVs, UGVs). Solid background in dynamical systems, control, estimation, and optimization, with experience translating advanced research into validated engineering solutions. Collaborative team player with international, interdisciplinary experience and strong communication skills.

Experience

Research & Development (PhD) *Dec 2020 - Jun 2025*
TU Delft, The Netherlands

- Developed advanced motion planning algorithms for ASVs; robustness to faults/uncertainties and safer navigation.
- Integrated ROS-based planning and control pipelines in simulation; automated experiments and performance analysis.
- Collaborated in multidisciplinary teams and published in international venues and journals (IEEE T-ITS, ECC).
- Supervised students on their MSc thesis.

Teaching Assistant *Nov 2019 - Jan 2020*
TU Delft, The Netherlands

- Supported MSc-level course on nonlinear systems theory, guiding and evaluating students in assignments.

Mechanical Engineering Intern *Jul 2016 - Oct 2016*
Maniatopoulos S.A. IDEAL BIKES, Patras, Greece

- Designed and prototyped components and tools for a cargo e-bike.
- Performed statistical quality analysis to reduce frame defects and improve manufacturing robustness.

Formula Student Team Member *Jul 2013 - Oct 2014*
Aristotle Racing Team, Thessaloniki, Greece

- Led design, structural analysis (CAD/CAE), and manufacturing of the vehicle's steel tubular frame.
- Coordinated subsystem integration, subsystem interfaces, assembly sequencing, rule-compliance, and validation.
- Ensured competition readiness; achieved top-10 placements internationally.

Education

PhD in Cognitive Robotics *Dec 2020 – Jun 2025*
TU Delft, The Netherlands

- Thesis: *"Rule-compliant and Fault-tolerant Motion Planning: With Application to Autonomous Surface Vehicles"*.
- Developed real-time algorithms for collision avoidance with rule compliance, fault diagnosis, fault reconfiguration, and robustness to uncertainties, resulting in safer and more reliable navigation compared to existing solutions.
- Collaborated within an international, interdisciplinary research team and co-supervised MSc students on projects related to autonomous navigation.

MSc in Systems and Control *Sep 2018 – Dec 2020*
TU Delft, The Netherlands GPA: 8.56/10 ***Cum Laude***

- Thesis: *"Distributed IDA-PBC for Nonholonomic Mechanical Systems"*. Designed a distributed passivity-based control algorithm for a fleet of heterogeneous, underactuated, and nonholonomic robots.
- Specialized in control theory, dynamical systems, optimization, and adaptive/predictive/robust control methods.
- Collaborated in small international teams to deliver high-quality results in demanding course projects.

M.Eng. in Mechanical Engineering (Integrated BSc & MSc equivalent) *Sep 2011 – Jul 2018*
Aristotle University of Thessaloniki, Greece GPA: 8.08/10

- Thesis: *"The Effect of Power Distribution Architectures and Torque Vectoring in Vehicle Dynamics"*. Investigated vehicle dynamics under different power distribution architectures and developed a torque-vectoring controller.
- Specialized in mechanical design, analysis, synthesis, and control of mechanical systems.
- Member of Formula Student team; collaborated to deliver a competitive racing vehicle for international competitions.

Publications (Selected)

- **Tsolakis, A.**, Negenborn, R., Reppa, V., Ferranti, L. *Model Predictive Trajectory Optimization and Control for Autonomous Surface Vessels Considering Traffic Rules.* IEEE Trans. Intell. Transp. Syst., 25(8):9895–9908.
- **Tsolakis, A.**, Ferranti, L., Reppa, V. *Set-Membership Estimation for Fault Diagnosis of Nonlinear Systems.* In *Proc. European Control Conference (ECC)*, Thessaloniki, Greece, 24–27 Jun. 2025.
- **Tsolakis, A.**, Benders, D., De Groot, O., Negenborn, R., Reppa, V., Ferranti, L. *COLREGs-aware Trajectory Optimization for Autonomous Surface Vessels.* IFAC-PapersOnLine 55(31):269–274.
- **Tsolakis, A.**, Keviczky, T. *Distributed IDA-PBC for a Class of Nonholonomic Mechanical Systems.* IFAC-PapersOnLine 54(14):275–280.

Awards & Distinctions

- MSc Systems & Control *Cum Laude*, TU Delft.
- Young Author Award nominations (IFAC-MICNON 2021, IFAC-CAMS 2022).
- Formula Student: 4th overall (FS Italy); 12th overall (FS Germany); top-10 in Engineering Design (FS Hungary).

Skills

- **Programming:** C++, Python, Matlab/Simulink, ROS2
- **Tools:** Git; CAD (SolidWorks, Inventor, Creo Parametric); FEA/CAE (ANSA, Altair MotionView)
- **Languages:** Greek (native), English (fluent), French (A1, learning)
- **Interests:** Rock climbing, hiking, skiing, sailing, scuba diving, motorcycle trips, classical guitar

Personal Projects

- **nav-mpc — Real-time Nonlinear MPC Framework for Autonomous Navigation**
  Nonlinear MPC · OSQP · Embedded Control · Python · ROS2
  Personal research project focused on designing a real-time nonlinear MPC framework for autonomous navigation, emphasizing computational efficiency, modularity, and suitability for embedded robotic platforms.
  - Symbolic formulation of nonlinear MPC problems (dynamics, constraints, objectives).
  - Automatic reformulation into a Linear Time-Varying QP with Cython-accelerated evaluation.
  - Real-time QP solution using OSQP, achieving sub-millisecond solution times.
  - Integrated simulation, plotting, and animation tools for rapid prototyping and validation.
  - Demonstrated on double pendulum full swing-up and mobile robots; designed for UGVs, UAVs, and USVs.
  - Intended as the core planning module for the Autonomous Rover and FLARE projects.
- **Autonomous Rover — ROS2-Based Mobile Robot with Full Onboard Autonomy**
  Rover · Raspberry Pi 5 · LiDAR · IMU · Odometry · ROS2
  Hands-on project to design and build a fully autonomous four-wheel differential-drive rover with complete onboard perception, planning, and control.
  - Designed a serviceable 3D-printed enclosure with custom mounting, cable management, and maintenance access.
  - Hardware/software bring-up, network configuration, logging, and analysis of field runs.
  - Sensor fusion of wheel odometry, IMU, and 2D LiDAR using an EKF.
  - Distributed architecture with separate onboard computers for SLAM/perception and planning/control.
  - Developed as an experimental platform for real-time MPC-based navigation.
- **FLARE — Solar-Powered UAV for Persistent Wildfire Surveillance**
  UAVs · Solar Power · Environmental Monitoring · Fire Detection
  Research-driven project to develop FLARE (Fire-Line Aerial Reconnaissance and Early Warning), a solar-powered, long-endurance UAV system for continuous wildfire monitoring.
  - Developed a wildfire spread simulator using cellular automata and GeoTIFF-based vegetation maps.
  - Conducted payload studies and preliminary airframe sizing and optimization for $> 24\,\mathrm{h}$ endurance.
  - Conceptual design of autonomy stack for a solar-powered autonomous glider with thermal and RGB payloads.
  - Following a model-based systems engineering (MBSE) approach using the V-model, stakeholder analysis, operational scenarios, functional analysis, and requirements definition.
  - Submitted as an MSCA Postdoctoral Fellowship (MSCA-PF) research proposal.